

# Timeouts retries and cancellation

Project: Grounded Learning Service

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of timeouts retries and cancellation.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of timeouts retries and cancellation

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Approval gates and idempotency. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

#### My starting explanation

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### This week's end goal

Explain timeouts retries and cancellation using the deciding step in this fictional service.

## 2. Notice the evidence

Timeouts bound waiting; retries consume a finite budget. Cancellation should stop future work while accounting for operations already committed.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is  $3 \cdot 4 + 1 + 2 = 15$  seconds.

### Timeouts retries and cancellation — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.

**Check or calculation:** Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is  $3 \cdot 4 + 1 + 2 = 15$  seconds.

**Conclusion and limit:** Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: Cancellation arrives after the first timeout.

Prediction: Do not start the second attempt.

Reason: Cancellation overrides the unused retry allowance.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Timeout

**Meaning:** A declared limit on waiting

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**Word:** Retry budget

**Meaning:** The maximum repeat attempts or resources

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**Word:** Cancellation

**Meaning:** A request to stop future processing

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.

Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
def waiting_bound(attempts, per_attempt, gaps):  
    if type(attempts) is not int or attempts < 1 or len(gaps) != attempts -  
1:  
        raise ValueError('attempt budget')  
    if per_attempt < 0 or any((g < 0 for g in gaps)):  
        raise ValueError('negative duration')  
    return attempts * per_attempt + sum(gaps)  
  
print(waiting_bound(3, 4, [1, 2]))
```

## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Timeouts retries and cancellation

Retry within declared limits. A read-only lookup allows two attempts of at most one second each. Cancellation forbids further attempts.

#### The facts we will use

First attempt times out at 1 second. Second succeeds in 0.4 seconds. Ignore overhead. No cancellation occurs.

AI ACADEMY / AI-6/16

## Timeouts retries and cancellation

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Bound first wait

Stop attempt one after its one-second allowance.



#### 2. Check retry

One attempt remains, and cancellation has not occurred.



#### 3. Complete next attempt

Accept the checked record; total waiting is 1.4 seconds.

### CHANGE ONE THING

Cancellation arrives after the first timeout.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Bound first wait

**What happens and why:** Stop attempt one after its one-second allowance.

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**Step:** 2. Check retry

**What happens and why:** One attempt remains, and cancellation has not occurred.

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**Step:** 3. Complete next attempt

**What happens and why:** Accept the checked record; total waiting is 1.4 seconds.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

A timeout bounds waiting; a retry tries again; cancellation withdraws the request. They are separate conditions on the next action.

### What this example does not establish

The arithmetic excludes overhead; retries of side-effecting operations require separate duplicate and recovery controls.

Read the labelled picture: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?

Change one thing: Cancellation arrives after the first timeout. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided:** restate the worked result, then name the assumption or limitation that prevents a

**broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Timeouts retries and cancellation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Timeouts retries and cancellation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

```
def waiting_bound(attempts, per_attempt, gaps):  
    if type(attempts) is not int or attempts < 1 or len(gaps) != attempts -  
1:  
        raise ValueError('attempt budget')  
    if per_attempt < 0 or any((g < 0 for g in gaps)):  
        raise ValueError('negative duration')  
    return attempts * per_attempt + sum(gaps)  
  
print(waiting_bound(3, 4, [1, 2]))
```

#### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

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On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. Explain how omitting it could change a reader’s interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled timeouts retries and cancellation evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of timeouts retries and cancellation. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

## Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

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## 13. Study a complete case

### Complete evidence record

#### Timeouts retries and cancellation — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.

**Check or calculation:** Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is  $3 \cdot 4 + 1 + 2 = 15$  seconds.

**Conclusion and limit:** Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

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The figure connects this chapter’s worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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## 14. Try a new case independently

## Independent case: Apply a new case independently

Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Attempt this without hints

Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

### My independent procedure, result and limitation

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: Retry within declared limits. A read-only lookup allows two attempts of at most one second each. Cancellation forbids further attempts.

### By the end, I can · How I will show it

**By the end, I can:** Explain the mechanism and terminology of timeouts retries and cancellation

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of timeouts retries and cancellation

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Retry within declared limits. A read-only lookup allows two attempts of at most one second each. Cancellation forbids further attempts.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept the checked record; total waiting is 1.4 seconds.

1. Complete next attempt
2. Bound first wait
3. Check retry

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Complete next attempt Accept the checked record; total waiting is 1.4 seconds. A timeout bounds waiting; a retry tries again; cancellation withdraws the request. They are separate conditions on the next action.

### Practice 2: explain the deciding evidence

Changed-case practice: Cancellation arrives after the first timeout. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Do not start the second attempt.
3. This one example guarantees correct results for every future input.

### **My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Do not start the second attempt. Cancellation overrides the unused retry allowance.

### **Repair a missing building block**

Review Approval gates and idempotency (ai-6/15). Goal: Explain the mechanism and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram. Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the mechanism and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

### **Predict, run and debug**

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.